Six-slide introduction

Slide 1: Python Learning Pathway

- First-year CSE • No programming background
- 28 days × 30 minutes = 14 hours
- Goal: explain, write and test small Python programs

Slide 2: Why this learning design?

- Official tutorial assumes prior programming experience
- Add plain-language examples and trace tables
- Daily retrieval and distributed practice
- Evidence: Python tutorial; Dunlosky et al. (2013)

Slide 3: Four-week roadmap

- Week 1: values, variables, input and strings
- Week 2: decisions, loops and boundaries
- Week 3: lists, dictionaries and functions
- Week 4: validation, tests and a score tracker

Slide 4: AI as a patient tutor

- Predict → attempt → one hint → run → explain
- Ask one question at a time
- Use errors as clues, not a reason to copy
- Protect personal information; credit assistance

Slide 5: Practice, tests and automation

- 20 exercises with reference solutions
- Run: python -m unittest discover -s tests -v
- Local .eml drafts generated from the daily plan
- No email sent; no API keys required

Slide 6: Demonstration and honest evaluation

- Run score tracker; show an invalid input
- Generate day 1 practice draft
- Collect real pre/post quiz scores; do not invent gains
- Complete four-tool evidence log before submission
- Sources and limitations: docs/research_report.md